

Human Development Index (HDI) Predictor: ML Web Application

Project Description

The Human Development Index (HDI) Predictor is a data-driven machine learning web application designed to evaluate and predict a country's development index based on three key socio-economic indicators defined by the United Nations: healthcare (Life Expectancy at Birth), education (Mean Years of Schooling), and economic output (Gross National Income per Capita).

Utilizing a pre-trained Linear Regression model trained on historical global indicators, the platform processes simulated socio-economic parameters and returns a precise predicted HDI score (0.000 to 1.000) along with its corresponding developmental tier (Low, Medium, High, Very High). Built on Python-Flask and featuring a modern glassmorphism user interface with country autocomplete features, the app allows researchers, students, and policy planners to test target interventions without needing local software or code execution.

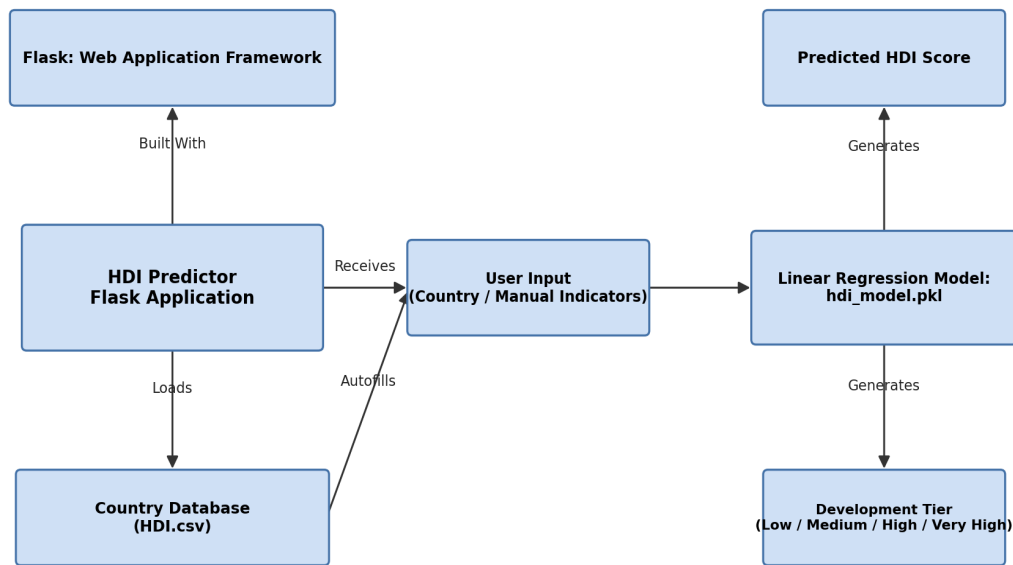
Scenarios

Scenario 1 (Policy Simulation): A policy planner selects a country (e.g., "Kenya") from the autocomplete search box. The system auto-fills the inputs with Kenya's latest historical indicators. The planner simulates an increase in mean years of schooling from 6.6 to 9.0 and views the projected HDI impact and tier progression in real-time.

Scenario 2 (Hypothetical Projections): A researcher inputs manual, customized socio-economic indicators (not tied to any specific country) to observe how a theoretical developmental profile maps to standard UNDP-aligned classification tiers.

Scenario 3 (Educational Analysis): A development economics student searches through different nations to analyze how baseline ratios of Life Expectancy, Schooling, and GNI behave and compare globally using the pre-loaded in-memory database.

Technical Architecture



Description: The HDI Predictor uses a modular architecture to deliver fast, ML-driven development index predictions. The frontend provides a glassmorphism form-based interface built with HTML, CSS, and vanilla JavaScript, while the Flask backend handles input parsing and passes the data directly to a pre-trained scikit-learn Linear Regression model. The model returns a predicted HDI score, which the backend maps to a development tier and renders back to the user. Country selection is powered by an in-memory database (HDI.csv) that auto-fills historical indicators upon selection.

(Note: The HDI Predictor does not use a live database for prediction storage – the trained model is loaded from a serialized .pkl file, and the country reference data is loaded from a bundled flat-file, so no ER Diagram is applicable.)

Pre-requisites

- **Flask Web Framework:** Flask Documentation (<https://flask.palletsprojects.com>)
- **scikit-learn / ML Libraries:** scikit-learn Documentation (<https://scikit-learn.org>)
- **HTML, CSS & JS Proficiency:** W3Schools Web Development Tutorials
- **Python Programming:** Python Documentation (<https://docs.python.org>)
- **Version Control & Deployment:** GitHub and Render Hosting Documentation

Project Workflow

Milestone 1: Model Development & Analytics

This milestone focuses on preparing the dataset, selecting a regression model, and serializing it for production use.

Activity 1.1: Prepare the Dataset

- Gather the Kaggle/UNDP historical indicator dataset containing columns for country, year, life expectancy, schooling, GNI, and final HDI.
- Clean the data and split it into training and test sets.

Activity 1.2: Research and Select the Regression Model

- Compare regression algorithms (Linear Regression, Ridge) in a Jupyter Notebook.
- Select the baseline Linear Regression model for its high R^2 score and fast inference.

Activity 1.3: Model Serialization

- Train the final model on the entire dataset.
- Export the serialized weights (hdi_model.pkl) using Pickle or Joblib.

Milestone 2: Backend Development (Flask/app.py)

Activity 2.1: Flask Server

- Set up the Python server and load the pre-trained hdi_model.pkl and country database HDI.csv in-memory at startup.

Activity 2.2: API Endpoint

- Program the /predict route to accept incoming JSON or form parameters, parse values into a NumPy matrix, run the regression model, and map the score to developmental tiers:
- *Low*: $HDI < 0.550$
- *Medium*: $0.550 \leq HDI < 0.700$
- *High*: $0.700 \leq HDI < 0.800$
- *Very High*: $HDI \geq 0.800$

Example tier-mapping logic used in the /predict route:

```
def get_tier(hdi_score):

    if hdi_score >= 0.800:

        return "Very High"

    elif hdi_score >= 0.700:

        return "High"

    elif hdi_score >= 0.550:

        return "Medium"

    else:

        return "Low"
```

Milestone 3: Frontend Development (templates/indexnew.html)

Activity 3.1: Glassmorphism Layout

- Design a single-page CSS dashboard with frosted glass elements, shadows, and smooth visual hierarchies.

Activity 3.2: Dropdown Autocomplete

- Write a vanilla JavaScript module that filters countries as the user types, pre-filling the inputs with corresponding indicators upon selection.

Activity 3.3: AJAX Integration

- Bind form submission to a JS event listener that fires asynchronous POST requests to the Flask server, preventing page reloads.

Milestone 4: Local Verification

Activity 4.1: Testing

- Build a local virtual environment (venv) and install dependencies via requirements.txt.
- Run the app locally and test predictions against known UNDP indices to confirm alignment.

```
python -m venv venv  
  
venv\Scripts\activate (Windows) or source venv/bin/activate (Mac/Linux)  
  
pip install -r requirements.txt  
  
python app.py
```

Milestone 5: Web Deployment

Activity 5.1: Hosting

- Push the codebase to a GitHub repository.
- Connect the repository to Render and configure build settings (pip install -r requirements.txt).
- Launch the production service and share the live URL.

Exploring the Website's Web Pages

Dashboard Input Page

HDI Predictor

Development Index Tool

Predict the Human Development Index

Search a country or enter indicators manually to estimate its HDI score.

Country Search

Life Expectancy at Birth (yrs)

Mean Years of Schooling

GNI per Capita (USD)

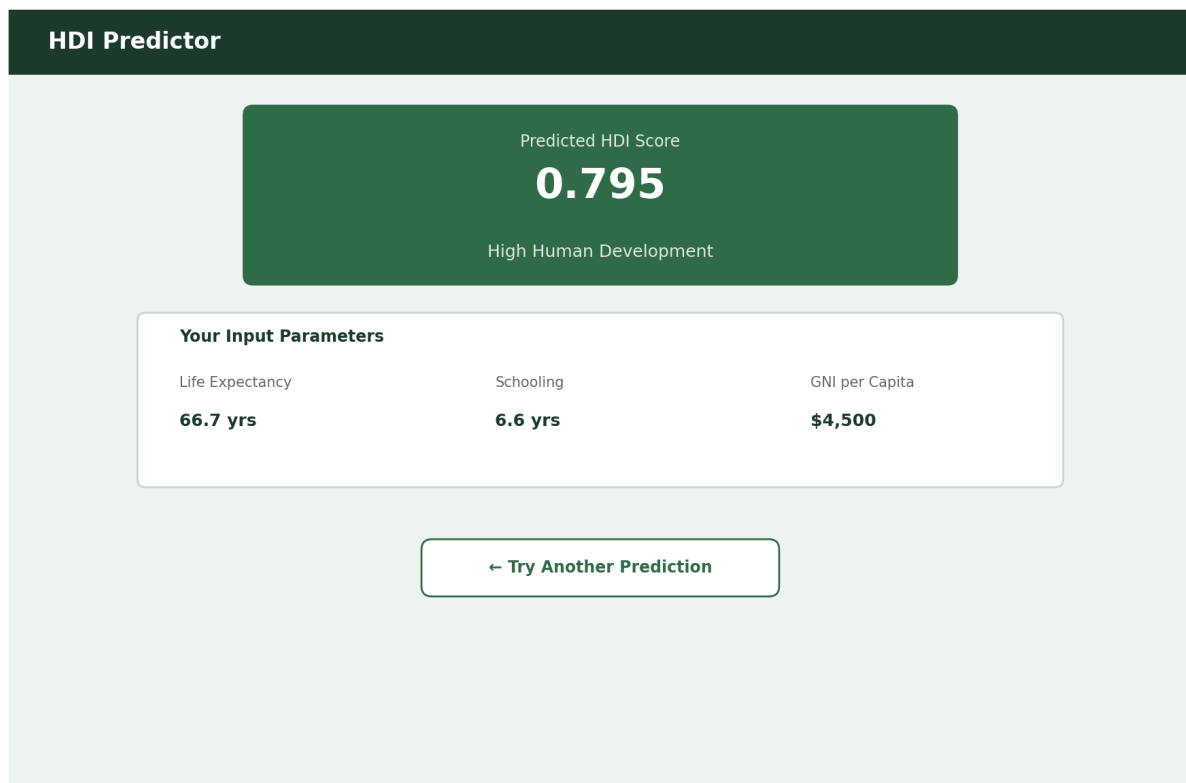
Predict HDI

HDI Predictor · Powered by Machine Learning · Built with Flask

Illustrative mockup of the dashboard input page

Description: The home page features a modern glassmorphism card containing a searchable country autocomplete input. Below the search box are three inputs (Life Expectancy, Schooling, GNI per Capita) with client-side boundary validators to block invalid ranges. A prominent "Predict HDI" submit button initiates calculations.

Results Dashboard



Illustrative mockup of the results dashboard

Description: Upon form submission, the UI displays the predicted index value (e.g. 0.795) in a styled gauge indicator. Below the score, a colored badge highlights the corresponding development category (e.g., "High Human Development") accompanied by a list of the input variables used in the query.

Conclusion

The HDI Predictor is a modular, lightweight machine learning web application that streamlines socio-economic target simulation. By coupling a serialized Linear Regression model with a responsive glassmorphism UI and a country autocomplete database, the application bridges the gap between complex statistical datasets and policy planners. Future expansions will focus on adding geographic maps and comparative side-by-side tracking grids.